

Radia

Carry luxury and comfort with you everywhere.

A collapsible, radial firepit that merges outdoor pursuits with modern comfort. Compact enough to fit in a rucksack, it offers multidirectional capabilities for shared warmth and atmosphere.

Discipline

Industrial Design

Context

UAL Foundation

Tools

Steel, Fusion, Blender

Outcome

Chosen for the UAL Millbank Exhibition — hand-formed and spot-welded 1:1 working steel prototype.

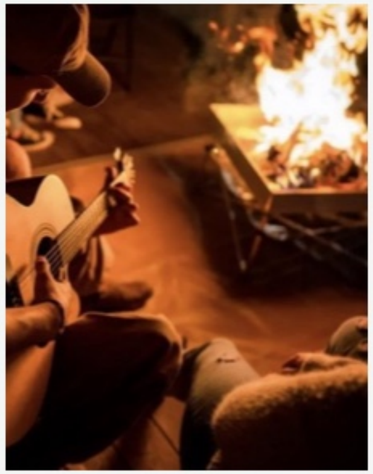
Research & insight



Primary research — camping trip



Solo Stove Ranger 2.0



Fireside Outdoor Pop-Up Pit



Cooker Portable Stove (Yangsheng Xia)



Elves Accessories (Qing Yan)



Portable Warmth Without the Bulk

Existing portable firepits force a compromise: double-walled cylinders like the Solo Stove offer clean combustion but cannot pack into a backpack, while mesh pop-up pits lack durability and radiant heat control.

Primary field observation during a multi-day camping trip showed that evening socialising revolves around a central heat source—driving the brief for a flat-packable, sheet-steel radial dish.

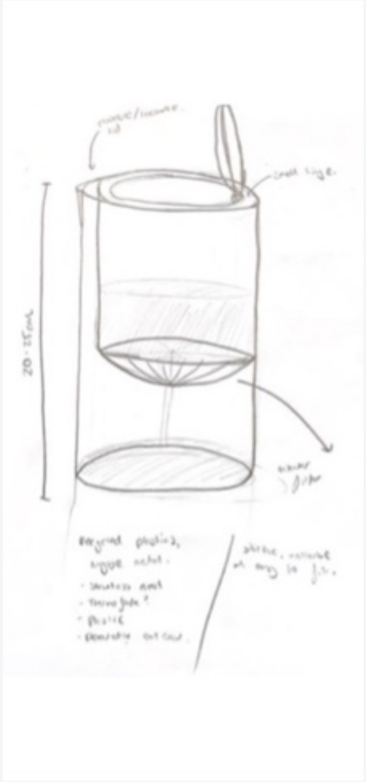
01 **Packability** — Must collapse into a single wedge stack that slides flat against a rucksack frame.

02 **Radiant Geometry** — Conical dish reflects heat upward and outward for shared warmth across a group.

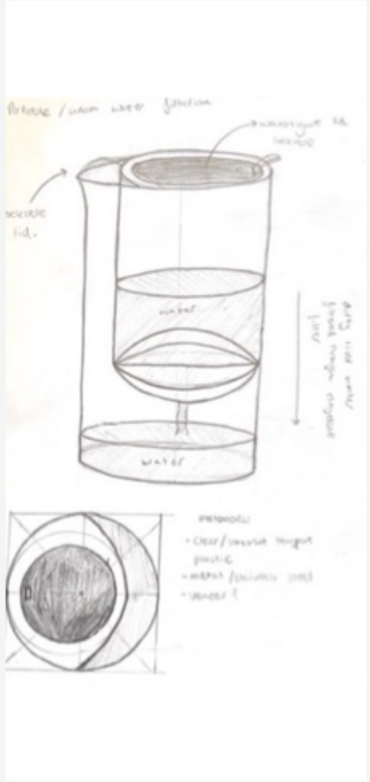
03 **Elevated Base** — Conical collar stand raises the fire dish off damp ground to protect grass and soil.

Ideation — from water filter to radial firepit

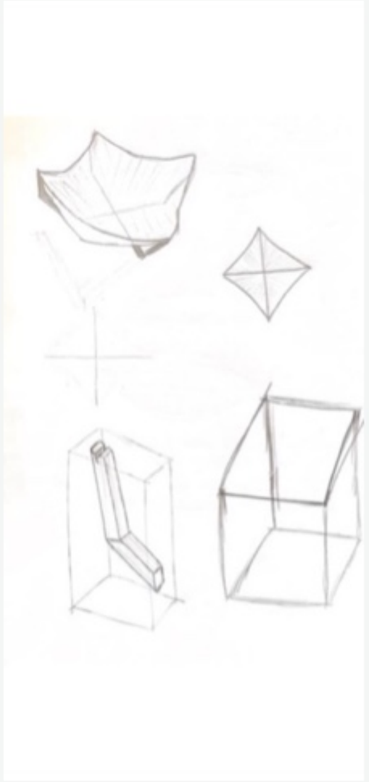
My initial concept explored a 'Portable Luxury Water Filter' before pivoting toward shared evening comfort at camp. Through six iterative sketch studies, I evolved the geometry from a rigid vessel into a 16-blade overlapping radial fan supported by a collapsible conical stand.



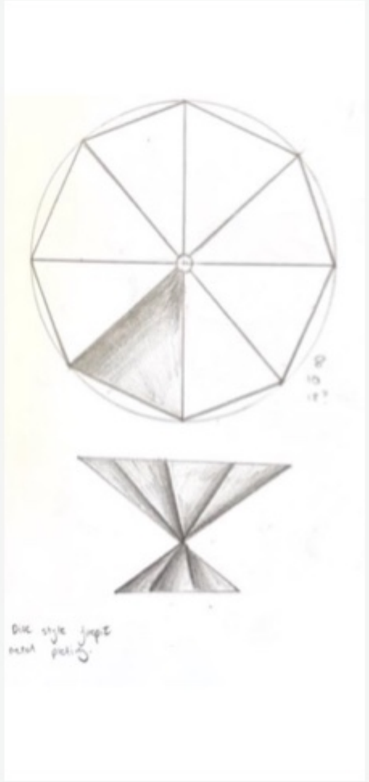
01 — Original 'Luxury Water Filter'



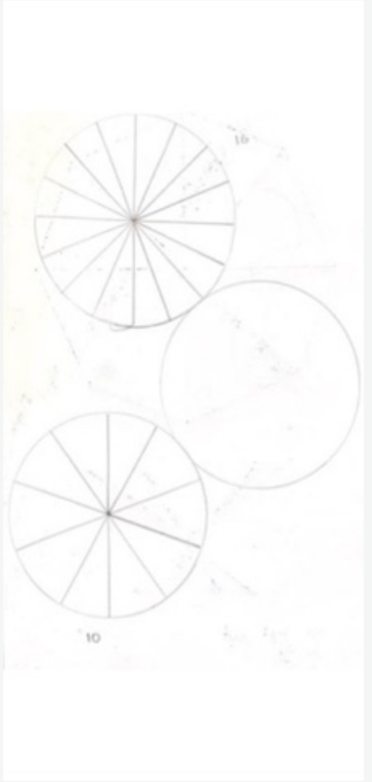
02 — Internal filter & vessel study



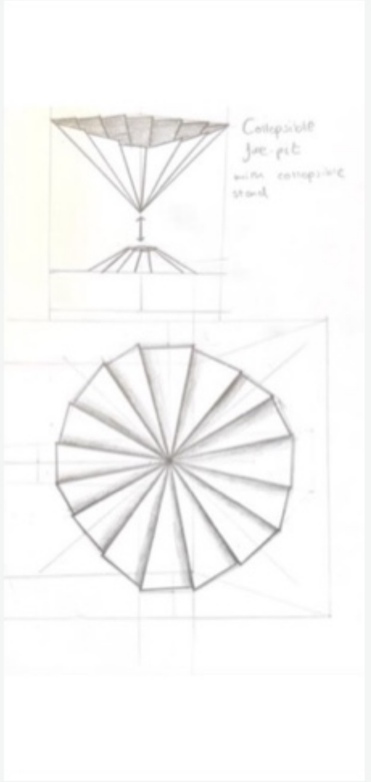
03 — Folding angular sheet forms



04 — Disc-style firepit & cone base

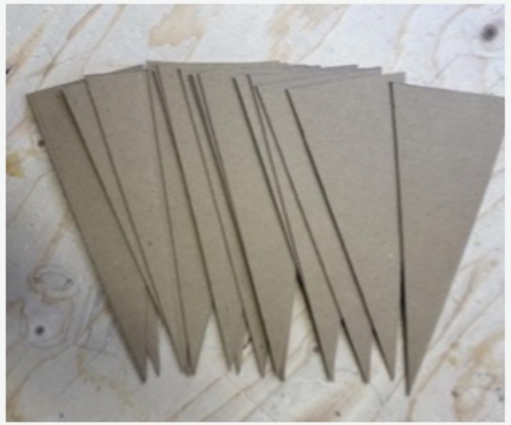


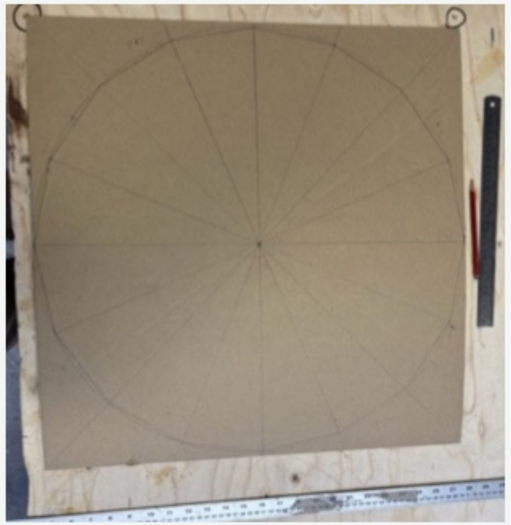
05 — 10 vs 16 radial blade layout



06 — Final collapsible radial firepit

Scale prototyping — 1:1 cardboard fire dish





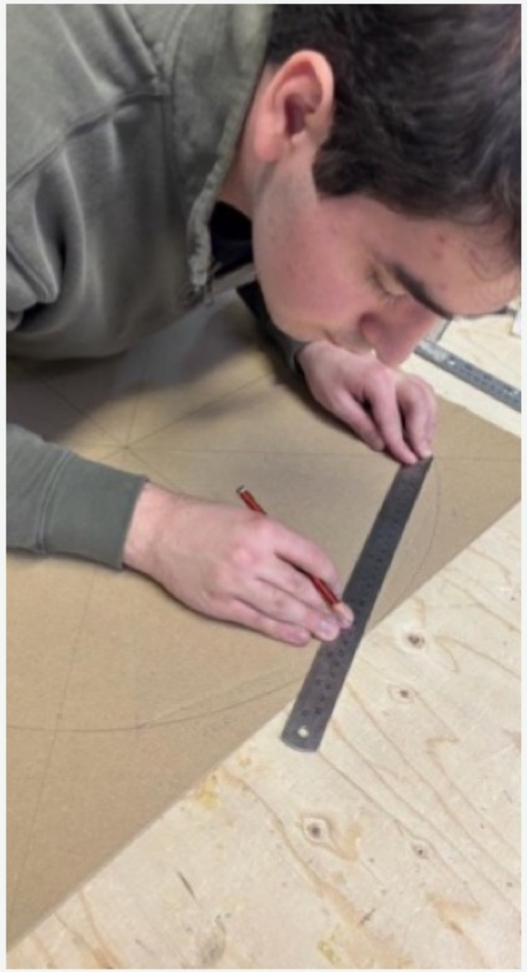


Fig. 01–03 — Marking out radial compass geometry on heavy card and cutting the 16 individual fan wedges



First Physical Iteration Findings

Testing the 1:1 cardboard dish with real timber offcuts confirmed the overlapping fan blades held fuel securely, while revealing that the slotted card tripod base needed to evolve into a rolled, riveted sheet-steel collar with a positive mechanical stop.

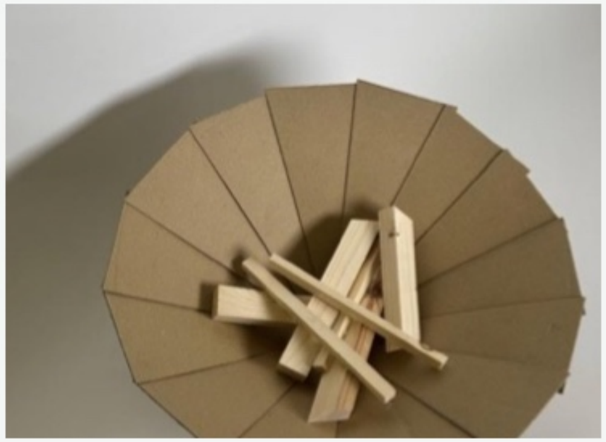
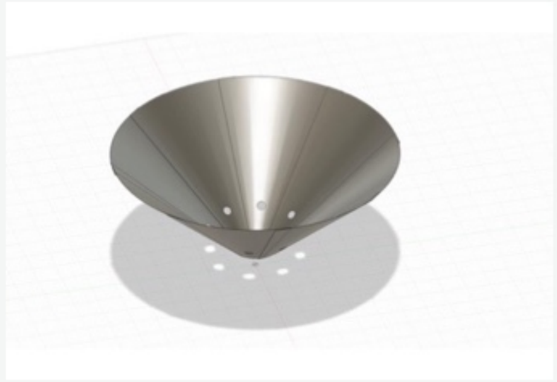
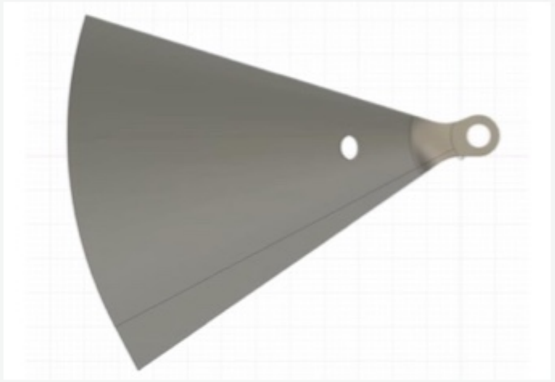


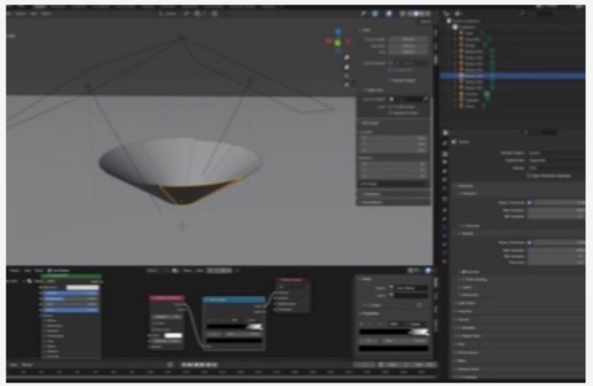


Fig. 04–06 — Testing log load and overlap lip retention

CAD development & metalworking process







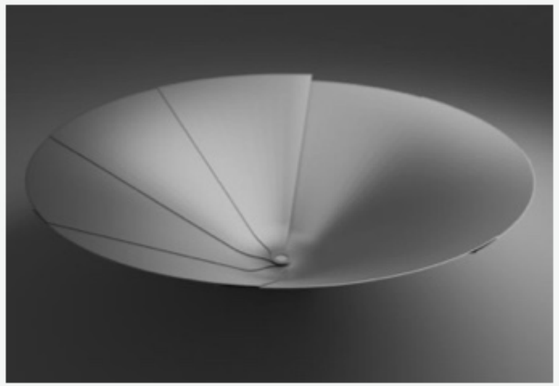


Fig. 07–10 — Fusion 360 sheet-metal cone & single blade with folded stop-lip, plus Blender studio visualisations



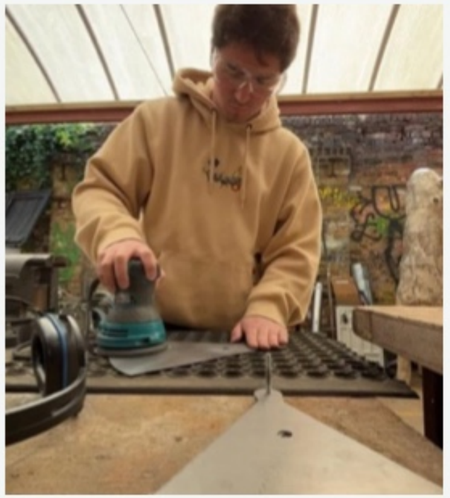
01 — Bandsaw profile cutting



02 — Deburring & edge filing



03 — Drilling ventilation & pivot holes

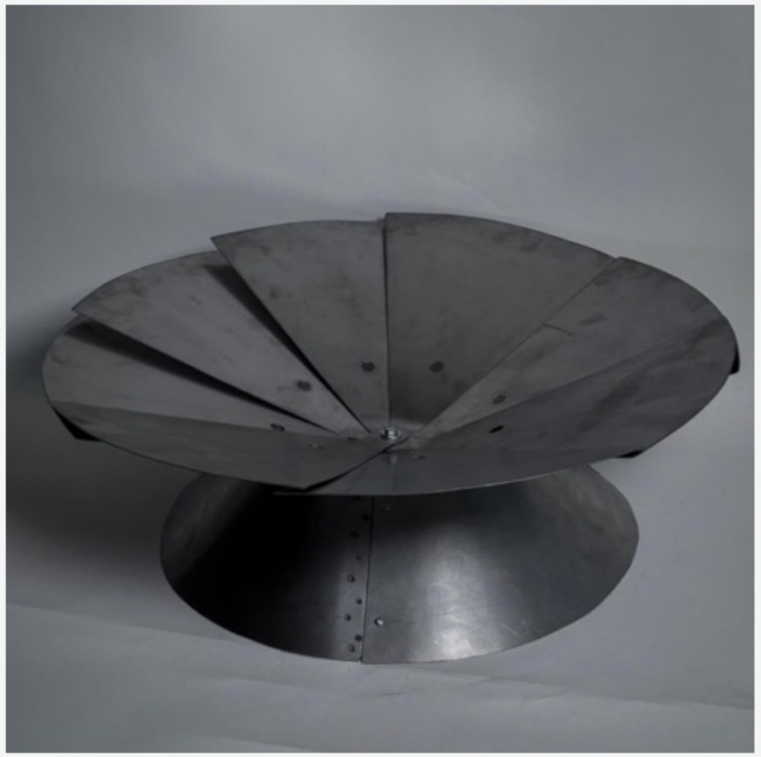


04 — Orbital surface finishing



05 — Spot-welding & riveting collar

Final design



01 — Deployed Firepit

Eight overlapping sheet-steel wedges fan out over a riveted conical base collar, elevating the fire above ground level.



02 — Radial Airflow Dish

Precision-drilled inner airflow holes feed oxygen to the base of the embers while folded blade edges lock the dish into a rigid cone.



03 — Flat-Pack Rucksack State

Unclipping the central bolt allows all wedges and the curved base collar to nest flat into a compact rucksack bundle.